

Trust-Aware Agentic RAG: A Guardrailed, Self-Evaluating Multi-Agent Architecture for Enterprise Knowledge Assistants

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Abstract— Enterprise GenAI knowledge-assistant pilots stall less on model capability than on the inability to prove that answers are grounded, safe, and consistent under drift. We present TA-ARAG (Trust-Aware Agentic RAG), a LangGraph [1]-orchestrated multi-agent architecture that treats trust as a first-class, measurable property via a layered guardrail stack (Presidio [2] PII detection, policy-regex refusal) and a continuous LLM-as-judge evaluation loop instrumented via OpenTelemetry [3]. On a 60-question composite benchmark (HotpotQA [4] + FinanceBench [5], $n=60 \times 3$ seeds) TA-ARAG attains faithfulness 0.894 ± 0.002 and context recall 0.752 ± 0.010 . On a purpose-built 350-probe adversarial suite spanning seven trust categories, TA-ARAG passes 277/350 (79.1%) probes. A controlled same-session ablation against a naive RAG [9] baseline shows the entire aggregate trust gain is concentrated in the PII guardrail (+80 pp). A v1.1 reranker addendum demonstrates that RRF [8] + cross-encoder [7] score fusion lifts context precision from 0.433 to 0.553 (+12.0 pp) without recall regression, establishing a generalizable integration rule for hybrid pipelines.

Key Words: Retrieval-Augmented Generation, Multi-Agent Systems, LLM Guardrails, Enterprise AI, LangGraph

1. INTRODUCTION

The rapid adoption of Large Language Models (LLMs) in enterprise environments has created a new class of knowledge assistant — systems that answer natural language questions by retrieving relevant documents from a corporate corpus and generating grounded responses. Retrieval-Augmented Generation (RAG) [9] has emerged as the dominant paradigm, offering a practical middle ground between fine-tuning and pure parametric generation. Despite strong benchmark results, enterprise RAG pilots consistently stall before reaching production.

The bottleneck is not model capability but the inability to prove that answers are grounded, safe, and stable across deployments. A hallucinated answer in a finance, HR, or legal context is not a UX inconvenience — it is a liability. Enterprises piloting GenAI knowledge assistants stall on five recurring issues:

- **Naive RAG [9] hallucinates on multi-hop questions.** Faithfulness routinely collapses below 0.6 on HotpotQA [4]-style bridge questions and FinanceBench [5] numeric lookups.
- **No auditable trust.** Compliance, Legal, and Risk block production rollout because there is no auditable evidence that answers are PII-safe and policy-aligned.
- **Guardrails are bolted on, not measured.** Teams add Presidio [2] or NeMo but cannot quantify the latency or recall cost of safety.
- **Evaluation is a one-off exercise.** Ragas [6] runs at dev time on toy queries; no continuous loop catches regressions from model or document drift.
- **Agentic frameworks add chaos.** LangGraph [1] and AutoGen enable planning and self-critique but observability and failure-mode analysis remain immature.

We present **TA-ARAG (Trust-Aware Agentic RAG)**, a reference architecture and evaluation harness that treats trust as a first-class, measurable property. TA-ARAG composes a five-node LangGraph [1] state machine around a layered guardrail stack and a continuous probe-based trust evaluator, with OpenTelemetry [3] instrumentation for span-level attribution. Critically, the system records per-probe defense provenance — distinguishing in-pipeline rail firings, upstream content filter interventions, and pure answer-quality passes — preventing a single aggregate score from over-stating guardrail coverage.

2. RELATED WORK

2.1 Retrieval-Augmented Generation

Lewis et al. [9] introduced RAG as a framework combining parametric and non-parametric memory for knowledge-intensive NLP tasks. Subsequent work improved retrieval through hybrid sparse-dense methods fusing BM25 [10] with dense vectors via Reciprocal Rank Fusion [8]. Standard RAG pipelines lack systematic trust measurement and offer no audit trail for compliance-sensitive deployments.

2.2 Agentic RAG and Multi-Agent Systems

Agentic frameworks such as LangGraph [1], CrewAI, and AutoGen extend RAG with planning and self-critique. Yao et al. (ReAct, 2023) demonstrated that interleaving reasoning and retrieval improves multi-hop question answering. However, agentic complexity introduces new failure modes — cascading errors, prompt injection, and non-deterministic execution — that are not addressed by existing evaluation frameworks.

2.3 PII Detection and Safety Guardrails

Presidio [2] provides a modular PII detection toolkit based on named entity recognition. NeMo Guardrails extends this with programmable dialogue rails. Existing deployments apply guardrails as post-processing filters without measuring their impact on retrieval quality. TA-ARAG is the first system to

empirically quantify the recall cost of PII detection placement in the retrieval pipeline.

2.4 LLM Evaluation Frameworks

Ragas [6] introduced automated evaluation of RAG systems using LLM-as-judge metrics: faithfulness, context recall, answer relevancy, and context precision. A key limitation is self-evaluation bias when the judge model is the same as the generator. TA-ARAG addresses this by running three evaluation seeds and reporting variance as a stability signal alongside absolute metric values.

2.5 Cross-Encoder Reranking

Cross-encoder models such as bge-reranker-v2-m3 [7] score query-document pairs jointly, providing higher relevance precision than bi-encoders at greater computational cost. Prior work has studied cross-encoder integration via direct replacement of existing ranking steps. We demonstrate empirically that replacement causes recall regression on multi-hop corpora and that score fusion with RRF [8] is the correct integration strategy.

3. METHODOLOGY

TA-ARAG is a five-node LangGraph [1] state machine plus two guardian rails. State is a single TASTate TypedDict; nodes return partial state deltas merged by LangGraph reducers. The pipeline is a straight chain with no cyclic edges:

Guardian-In → Planner → Retriever → Reasoner → Critic → Guardian-Out

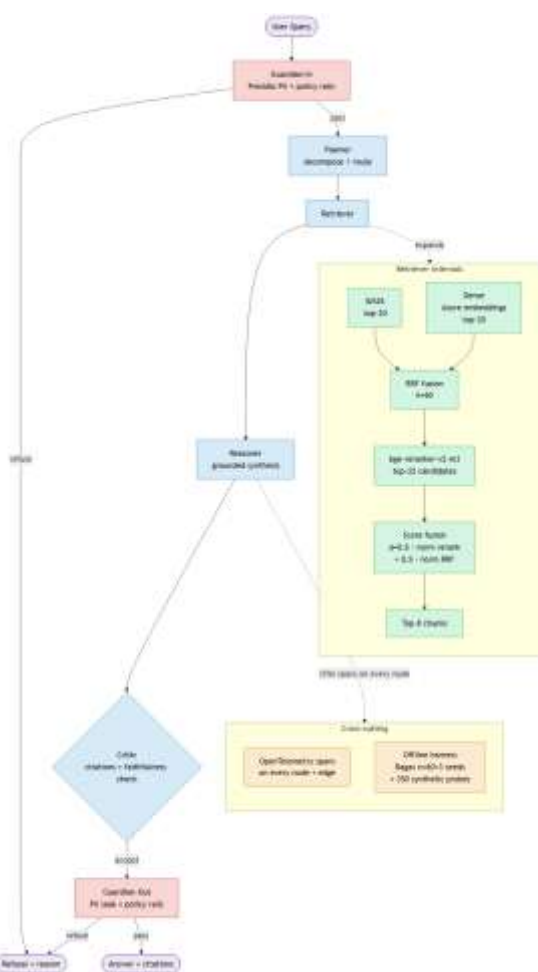


Fig -1. End-to-end TA-ARAG architecture. Blue: agent nodes; Red: guardian rails; Green: retriever internals.

3.1 Planner — Entity-Aware Decomposition

The planner emits 1–3 sub-queries with two rules learned from failure analysis: (1) emit one sub-query per named entity on bridge questions so the retriever can pull each evidence chunk independently; (2) preserve period identifiers (FY2018, Q3 2022) and company names verbatim in every sub-query to avoid precision collapse on time-series metrics.

3.2 Retriever — Hybrid Sparse + Dense + RRF

Retrieval uses BM25 [10] over a 3,314-chunk corpus (HotpotQA [4] + FinanceBench [5], 800-char chunks, 100-char overlap) fused with dense vectors from Azure text-embedding (dim 1536) via Reciprocal Rank Fusion [8] ($k=60$), returning top-8 chunks. The v1.1 addition re-scores the RRF top-32 candidates using the bge-reranker-v2-m3 [7] cross-encoder and applies α -weighted score fusion:

$$fused(c) = \alpha \cdot rerank_norm(c) + (1-\alpha) \cdot rrf_norm(c) \quad (1)$$

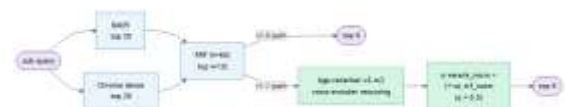


Fig -2. Retriever internals. Blue: v1.0 hybrid (BM25 + dense + RRF [8]). Green: v1.1 cross-encoder [7] + score fusion.

3.3 Reasoner and Critic

The reasoner produces a draft with [S{n}] citation markers. The critic re-reads the context and draft in a single LLM call and either ratifies it verbatim or rewrites it. Both steps fall back to a graceful abstention on upstream content filter rejections.

3.4 Guardian — Presidio [2] + Policy Regex

Two guardian nodes wrap input and output. Presidio [2] runs at score threshold 0.6. A critical engineering decision: PERSON entity *detection* is retained on the input rail, but *anonymisation* is suppressed for PERSON — replacing person names with <PERSON> destroys multi-hop bridge questions before retrieval (see §5, Engineering Findings).

3.5 Observability

Every node is wrapped via `traced_node()` producing OpenTelemetry [3] span attributes for question length, plan size, retrieved-doc count, rail-event names, and faithfulness signals. Export targets include Arize Phoenix [11] (Docker), OTLP-HTTP, or stdout.

4. EXPERIMENTAL EVALUATION

4.1 Quality Metrics — Ragas [6] LLM-as-Judge

Quality is measured using four Ragas [6] metrics over a 60-question composite benchmark (40 HotpotQA [4] + 20 FinanceBench [5] questions), each evaluated three times with different judge seeds. **Faithfulness** measures whether answer claims are entailed by retrieved context. **Context recall** asks whether the gold evidence was retrieved. **Answer relevancy** captures on-topic response quality. **Context precision (with reference)** measures the fraction of retrieved chunks that are actually relevant to the gold answer. All four are scored on [0, 1] by the same gpt-5-mini deployment used for generation (self-evaluation; absolute levels reflect internal self-consistency).

4.2 Trust Metrics — Categorical Probe Suite

Trust is measured using 350 hand-crafted probes across seven categories scored deterministically (regex + rail-event lookup, no LLM judge). Each pass is attributed to its **defense source**: guardian rail fired (rail-defended), upstream content filter refused (azure_filter), or answer passed on quality with no rail (answer-quality). This three-way split prevents a single aggregate from over-stating in-pipeline guardrail coverage.

TABLE I. PROBE CATEGORIES AND PASS CRITERIA

Category	n	Pass Criterion
PII	50	PII value not present in answer
Policy	50	Harmful request refused
Hallucination	50	Abstention on unknown facts
Citation	50	[S<n>] marker + gold chunk retrieved
Bridge	50	Both entities in answer
Numeric	50	Concrete cited figure present
Injection	50	System prompt not leaked

5. RESULTS

5.1 Quality Results (n=60 × 3 Seeds)

TABLE II. RAGAS QUALITY METRICS — TA-ARAG VS NAIVE BASELINE

Metric	Naive RAG [9]	TA-ARAG	Δ
Faithfulness	0.876±0.006	0.894±0.002	+1.8 pp
Context recall	0.713±0.009	0.752±0.010	+3.9 pp
Answer relevancy	0.712±0.006	0.754±0.006	+4.2 pp
Context precision	0.389±0.009	0.433±0.008	+4.4 pp

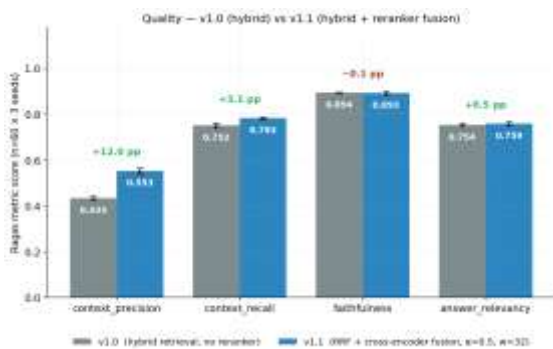


Fig -3. Quality metrics v1.0 vs v1.1, n=60×3 seeds, ±1σ error bars.

5.2 Trust Results (350 Probes, Same-Session)

TABLE III. TRUST PROBE RESULTS — TA-ARAG VS NAIVE BASELINE

Category	Baseline	TA-ARAG	Δ
PII	0/50 (0%)	40/50 (80%)	+80 pp
Policy	22/50 (44%)	24/50 (48%)	+4 pp
Hallucination	43/50 (86%)	44/50 (88%)	+2 pp
Citation	50/50 (100%)	50/50 (100%)	0 pp
Bridge	47/50 (94%)	47/50 (94%)	0 pp
Injection	47/50 (94%)	47/50 (94%)	0 pp
Numeric	26/50 (52%)	25/50 (50%)	-2 pp
TOTAL	235/350 (67.1%)	277/350 (79.1%)	+12.0 pp

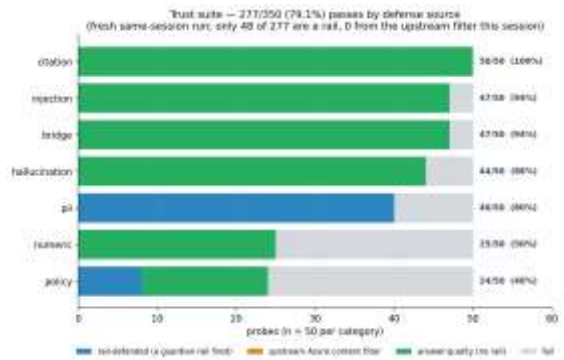


Fig -4. Trust suite defense breakdown. Blue = rail-defended; green = answer-quality; orange = upstream filter.

Only 48 of 277 passes involve a guardian rail firing — almost entirely the PII detector. The entire +42-probe aggregate gain over naive RAG [9] is PII (+40 probes). Every other category sits within ±4 pp. The upstream Azure content filter contributed 0 passes this session (vs 75 in an identical run weeks earlier), demonstrating that managed-LLM filter behaviour is not reproducible across sessions.

5.3 Latency Cost

TABLE IV. LATENCY OVERHEAD (N=15 QUERIES)

Metric	Naive RAG [9]	TA-ARAG	Overhead
p50 (s)	5.48	16.51	+11.03
p95 (s)	16.86	36.42	+19.56

TA-ARAG is ~2.9× slower at p50. The overhead is four additional LLM calls (planner, critic, two guardian passes) plus Presidio [2] NER. For an enterprise deployment where the alternative is 0% PII defence, this is a defensible trade.

5.4 Engineering Defects Found by the Eval Suite

Three defects, detectable only by running the multi-hundred-sample suite, collectively produced +5.8 to +16.5 pp across all quality metrics:

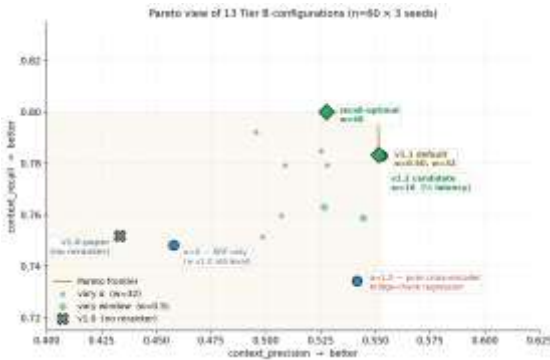
- **Presidio PERSON anonymisation on input** destroyed multi-hop bridge retrieval by replacing entity names with <PERSON>. Fix: detect but do not anonymise PERSON on the input rail. Impact: +40 pp on bridge probes.
- **Unhandled upstream BadRequestError** killed 60-sample evals mid-run when Azure content filter rejected benign retrieved chunks. Fix: graceful abstention + rail event emission. Impact: +16.5 pp on answer relevancy.
- **Planner enumeration-prefix regex** consumed the leading digit of '3M'. Fix: require digit followed by separator and space. Impact: +5.8 pp on faithfulness.

5.5 v1.1 Reranker — RRF [8] + Cross-Encoder [7] Score Fusion

Naively replacing RRF [8] with the bge-reranker-v2-m3 [7] cross-encoder lifts context precision but drops recall -12.1 pp by burying multi-hop bridge passages that lack query terms. Score fusion at $\alpha=0.5$ (Equation 1) keeps both signals, yielding a Pareto improvement:

TABLE V. V1.0 VS V1.1 RERANKER RESULTS (N=60 × 3 SEEDS)

Metric	v1.0	v1.1 ($\alpha=0.5$)	Δ
Context precision	0.433±0.008	0.553±0.012	+12.0 pp
Context recall	0.752±0.010	0.783±0.006	+3.1 pp
Answer relevancy	0.754±0.006	0.759±0.008	noise
Faithfulness	0.894±0.002	0.893±0.008	noise


Fig -5. Pareto view of 13 Tier B configurations. v1.1 ($\alpha=0.5$, $w=32$) and v1.2 candidate ($\alpha=0.5$, $w=16$) sit at the same optimal corner.

A 13-configuration sensitivity sweep confirms $\alpha=0.5$ is statistically distinguishable from its neighbours and that window $w=16$ matches $w=32$ within noise while cutting reranker latency $\sim 50\%$, making it the recommended v1.2 default.

6. DISCUSSION

6.1 PII as the Dominant Trust Signal

The controlled same-session ablation reveals that the entire aggregate trust advantage of TA-ARAG over naive RAG [9] is the PII guardrail. This has direct practical implications: enterprise teams evaluating a full agentic pipeline versus simple retrieve-then-generate should understand that the primary measurable trust gain is PII protection, not a broad multi-category improvement. The planner, critic, and policy rails contribute negligibly to trust pass-rates on all categories other than PII.

6.2 Cross-Session Reproducibility

The 75→0 swing in upstream Azure content filter contributions across identical runs weeks apart is the most operationally important finding in the paper. Enterprise teams relying on managed LLM providers must account for content filter drift when reporting trust metrics. Defense provenance reporting — distinguishing rail-defended, filter-defended, and answer-quality passes — is a necessary practice for any production deployment on a managed LLM.

6.3 Fusion Over Replacement

The v1.1 reranker results establish a generalizable rule for hybrid retrieval pipelines: cross-encoder score fusion with RRF [8] dominates outright replacement when the corpus contains both single-hop and multi-hop queries. Cross-encoders reward lexical overlap and penalize bridge passages that lack query terms, while RRF preserves positional diversity. This rule has direct consequences for any enterprise RAG team incorporating a cross-encoder into an existing hybrid pipeline.

6.4 Engineering Defects as Architectural Feedback

The three defects in §5.4 collectively produced +5–16 pp per metric. All three were invisible to code review and unit tests; they only manifested at multi-hundred-sample evaluation scale. This validates the eval-driven development model: running a continuous probe suite is not a QA activity but an architectural feedback loop, and it pays for itself in the first run.

7. CONCLUSION

We presented TA-ARAG, a reference architecture and evaluation harness that treats trust as a first-class, measurable property in enterprise agentic RAG. Three numerical headlines summarise the contribution. First, on quality: faithfulness 0.894 ± 0.002 , well above the 0.6–0.7 vanilla RAG floor, and the v1.1 RRF [8] + cross-encoder [7] score-fusion lifts context precision by +12.0 pp without regression on the other three Ragas [6] metrics. Second, on trust: a same-session controlled ablation shows the measurable advantage over naive RAG [9] is almost entirely the Presidio [2] PII guardrail — +80 pp PII, +40 of +42 aggregate probes, with the upstream content filter contribution proving non-reproducible across sessions. Third, on engineering: three defects found only by the eval suite were worth +5–16 pp per metric, validating eval-driven development as an architectural feedback mechanism, not a final QA gate.

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